One minute demo (5 slides)

Fast and Efficient Computation of Redundancy (KL minimax): From information geometry to superposed dualities in Laguerre geometry

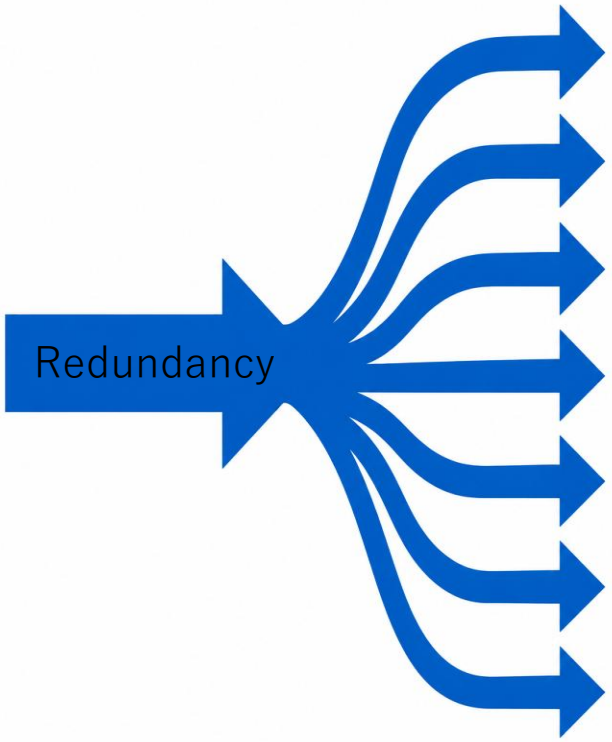


Redundancy: Minimize maximum error wrt to relative entropy:

$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\mathrm{KL}}(P : Q)$$

$$D_{\mathrm{KL}}(P : Q) = E_P[\log \frac{1}{Q(X)}] - H(P)$$

$$\mathcal{Q} = M_1^+(\mathcal{X}) \quad \mathcal{P} = \{P_1, \ldots, P_n\} \subset M_1^+(\mathcal{X})$$



- universal compression (source coding)
- online learning (sequential prediction)
- hypothesis testing
- minimum description length principle
- gambling
- model integration/fusion
- etc.

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When distributions belong to a same exponential family with cumulant function F

Example: Gaussians, categorical, Beta, Wishart: $p_\theta(y) = \exp(y \cdot \theta - F(\theta))$

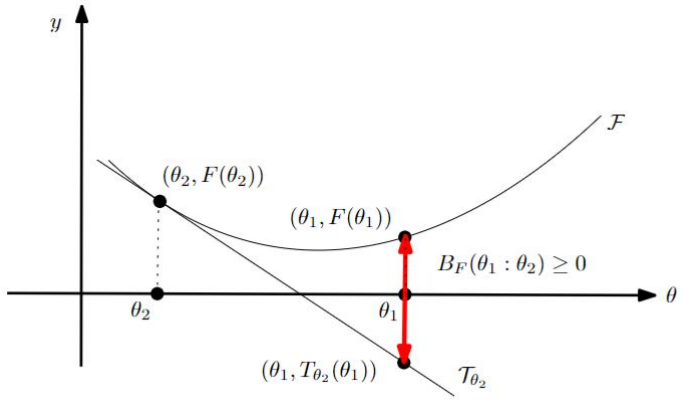
Right-sided KLD minimization

Left-sided BD minimization

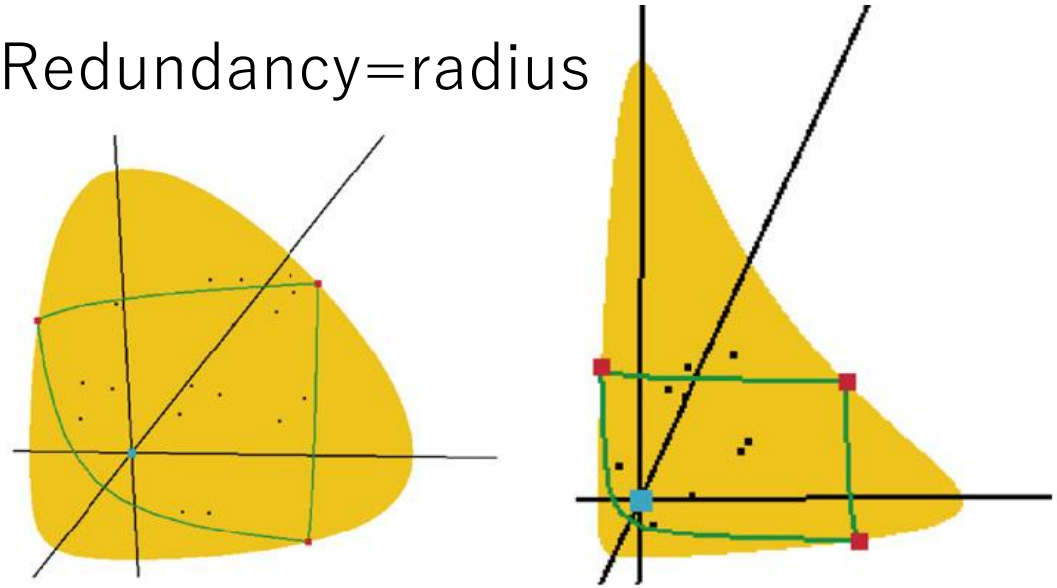
$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\mathrm{KL}}(P : Q) \quad \Rightarrow \quad \min_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

Bregman divergence

$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - < \theta_1 - \theta_2 , \nabla F(\theta_2) >$



Redundancy=radius



Kullback-Leibler Itakura-Saito

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One minute demo (5 slides)

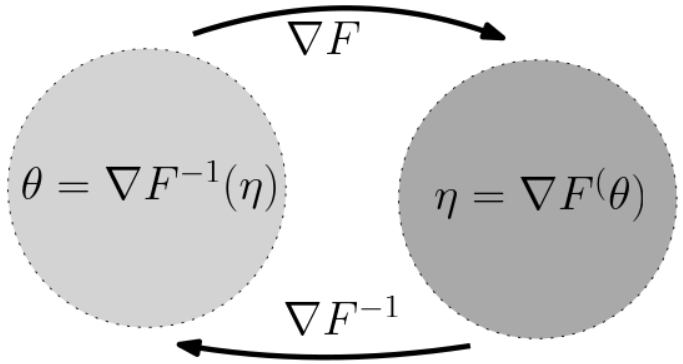
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Bregman divergence is rewritten as a power distance:

$$B_F(\theta_i : \theta_j) = \frac{1}{2}\,\sigma(\hat{\theta}_i, \check{\eta}_j)$$

$$\hat{\theta} = (\theta, \|\theta\|^2 - 2\,F(\theta))$$

$$\check{\eta} = (\eta, \|\eta\|^2 - 2\,F^*(\eta))$$



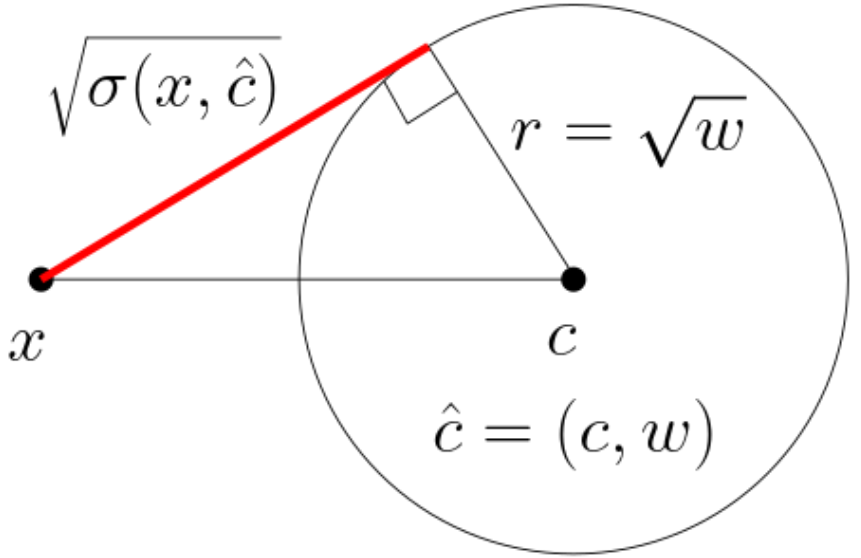
where F* is the Legendre-Fenchel transform of F:

$F^*(\eta) = \sup_{\theta \in \Theta}\{< \theta, \eta > - F(\theta)\}$

$\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta) = \nabla F^{-1}(\eta)$

Power distance between weighted points :

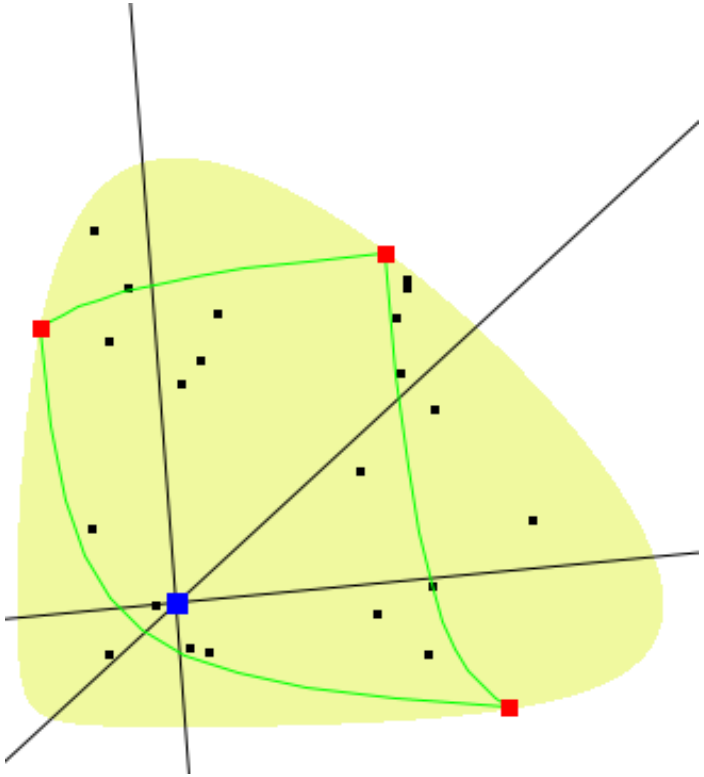
$$\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$$



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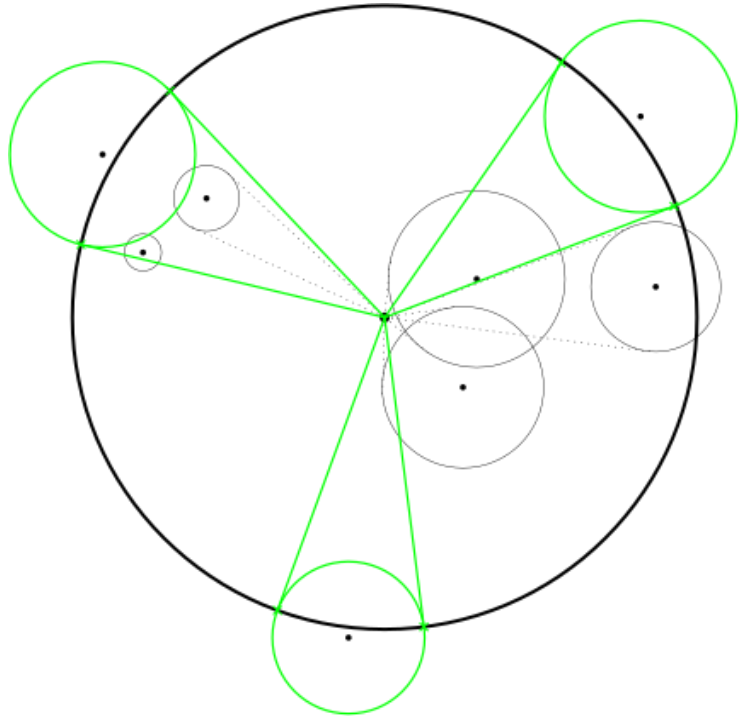
One minute demo (5 slides)

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$$B_F(\theta : \theta') = \frac{1}{2}\,\sigma(\hat{\theta} : \check{\eta}')$$

$$\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$$



Bregman miniball in θ

Power miniball in η

The key advantage of power miniballs is to be *independent of F*

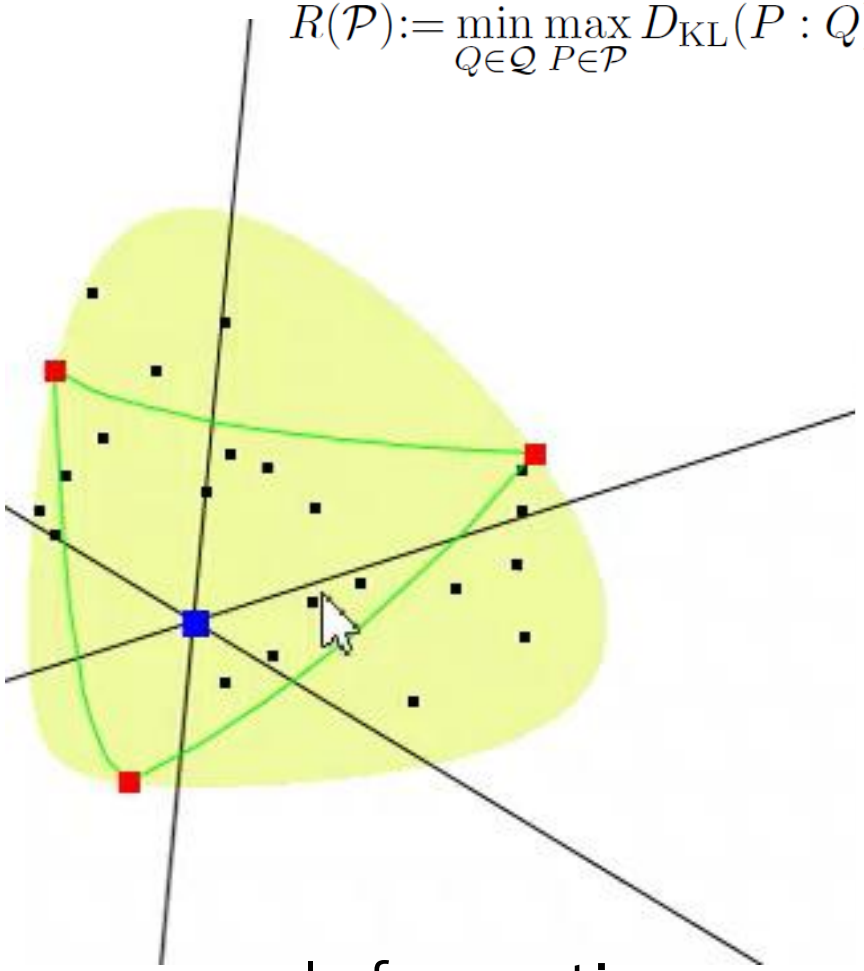
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Fast and Efficient Computation of Redundancy (KL minimax): From information geometry to superposed dualities in Laguerre geometry

Bregman enclosing ball

$$R(\mathcal{P}) := \min_{Q \in \mathcal{Q}} \max_{P \in \mathcal{P}} D_{\mathrm{KL}}(P : Q)$$

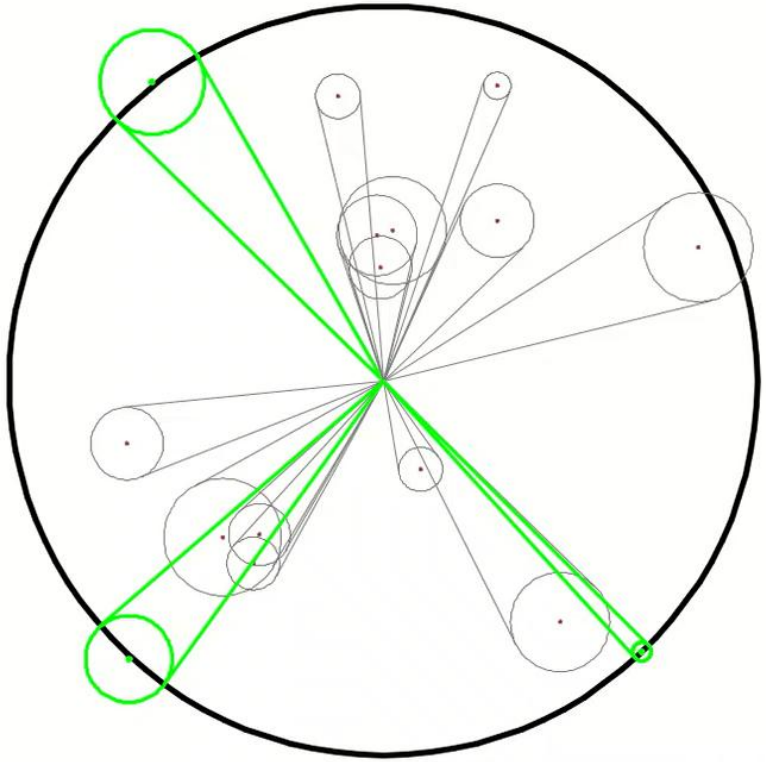


Information geometry

$$B_F(\theta : \theta') = \frac{1}{2}\sigma(\hat{\theta} : \check{\eta}')$$

$$\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$$

Power enclosing ball = orthosphere



Laguerre geometry

$$\min_p \max_i \sigma(p : \hat{p}_i)$$

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